

Section : Physics

- Q.1** A 0.5 m long solenoid has 500 turns and has a flux density of 2.52×10^{-3} T at its centre. The current flowing in the solenoid would be
(given, $\mu_0 = 4\pi \times 10^{-7}$ Hm $^{-1}$)
1. 2.0 A
 2. 3.0 A
 3. 4.0 A
 4. 5.0 A

Options 1. 1
2. 2
3. 3
4. 4

- Q.2** A sample of radioactive nuclei has a half life of 5.5 years. The fraction of sample remaining undecayed after 22 years is
1. 12.5%
 2. 25 %
 3. 93.75%
 4. 6.25%

Options 1. 1
2. 2
3. 3
4. 4

- Q.3** A bar magnet is released from rest along the axis of a very long, vertical copper tube. After some time, the magnet will
1. stop in the tube
 2. move with an acceleration equal to acceleration due to gravity
 3. oscillate
 4. move with almost constant speed

Options 1. 1
2. 2
3. 3
4. 4

- Q.4** The ground state energy of hydrogen atom is -13.6 eV. The kinetic and potential energies of the electron in this state will be
1. Kinetic energy of the electron = 13.6 eV; potential energy of the electron = -27.2 eV
 2. Kinetic energy of the electron = -13.6 eV; potential energy of the electron = 27.2 eV
 3. Kinetic energy of the electron = 13.6 eV; potential energy of the electron = -13.6 eV
 4. Kinetic energy of the electron = -13.6 eV; potential energy of the electron = -13.6 eV

Options 1. 1
2. 2
3. 3
4. 4

- Q.5** A parallel plate capacitor with plates of area A, held a distance d apart, has a capacitance C. A thin silver foil of negligible thickness and the same area is inserted at mid-point between the plates. The capacitance becomes
1. C/4
 2. 3C/4
 3. C/2
 4. C

Options 1. 1
2. 2
3. 3
4. 4

- Q.6** In a series L, C and R circuit with an ac source of frequency ω , the current leads the voltage by $\pi/4$. The value of capacitance C will be
1. $\omega(\omega L - R)^{-1}$
 2. $\omega^{-1}(\omega L - R)$
 3. $(\omega^2 L)^{-1}$
 4. $(\omega^2 L + \omega R)^{-1}$

Options 1. 1
2. 2
3. 3
4. 4

- Q.7** A converging lens and a diverging lens of equal focal lengths are placed coaxially in contact. The focal length and power of this combination will be
1. Focal length (F_c) = 0; power of combination (P_c) = 0
 2. Focal length (F_c) = ∞ ; power of combination (P_c) = 0
 3. Focal length (F_c) = 0; power of combination (P_c) = ∞
 4. Focal length (F_c) = ∞ ; power of combination (P_c) = ∞

Options 1. 1
2. 2
3. 3
4. 4

Q.8 Which of the following statements is correct for an electric dipole placed along the direction of a non-uniform electric field?

1. $F_{\text{net}} = 0$; $\tau_{\text{net}} = 0$
2. $F_{\text{net}} \neq 0$; $\tau_{\text{net}} = 0$
3. $F_{\text{net}} \neq 0$; $\tau_{\text{net}} \neq 0$
4. $F_{\text{net}} = 0$; $\tau_{\text{net}} \neq 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Transverse wave nature of light is shown by which phenomenon?

1. Reflection
2. Polarization
3. Interference
4. Diffraction

Options 1. 1

2. 2
3. 3
4. 4

Q.10 A proton is projected with a speed of $v = 4 \times 10^6 \text{ m s}^{-1}$ horizontally from east to west. A uniform magnetic field $\vec{B}$ of strength $B = 1.5 \times 10^{-3} \text{ T}$ exists in the vertically upward direction. What is the acceleration produced? (Use charge on the proton $q = 1.6 \times 10^{-19} \text{ C}$ and mass of the proton $m = 1.67 \times 10^{-27} \text{ kg}$)

1. $1.5.75 \times 10^{11} \text{ m s}^{-2}$
2. $2.5.8 \times 10^{10} \text{ m s}^{-2}$
3. $3.9.6 \times 10^2 \text{ m s}^{-2}$
4. $4.5.8 \times 10^{-11} \text{ m s}^{-2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.11 In reference to the radio waves, choose the **incorrect** statements:

- (A) Radio waves are the longest wavelength electromagnetic waves.
- (B) Radio waves are deviated by electric and magnetic fields.
- (C) Radio waves are diffracted easily.
- (D) Radio waves are not electromagnetic waves.

Choose the **correct** answer from the options given below:

1. (A) and (D) only
2. (C) and (D) only
3. (A) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.12 All the hydrogen-like atoms in their ground states have the same

1. orbital angular momentum of the electron
2. energy of the atom
3. speed of the electron
4. radius of the orbit

Options 1. 1

2. 2
3. 3
4. 4

Q.13 Which of the following statements are correct?

- (A) Electrostatic lines of force may be curved or straight in an electric field.
- (B) Number of lines of force per unit perpendicular area represents the intensity of the electric field at the point
- (C) Two electric field lines can not intersect each other.
- (D) Electric field lines are directed from negative to positive charge for an electric dipole.

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (C) and (D) only
3. (A), (B) and (C) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.14 There are two thin parallel charged sheets having surface charge density σ placed at a small distance apart. Which of the following statements are correct?

- (A) If both the sheets are positively charged, then the electric field between them is σ/ϵ_0
- (B) If both the sheets are negatively charged, then the electric field between them is σ/ϵ_0
- (C) If the left sheet is positively charged and the right sheet is negatively charged the field between them is σ/ϵ_0
- (D) If the right sheet is positively charged and the left sheet is negatively charged then the field between them is σ/ϵ_0

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (D) only
3. (B) and (C) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.15 In an inductive circuit, the current falls from 7.5 A to 1.5 A in 0.15 s. It induces emf of 220 V in the inductive coil. The inductance of the circuit is

1. 11 H
2. 5.5 H
3. 4.5 H
4. 3.9 H

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Which of the following statements are correct?

- (A) In dispersion through a prism of glass, the red color ray deviates most.
- (B) In simple microscope, to get a magnified image, the object is placed within the focus of a convex lens.
- (C) In compound microscope, total magnification is $m = (\text{magnification due to eye lens}) \times (\text{magnification due to objective})$
- (D) magnifying power of telescope $M = \text{focal length of eye lens} / \text{focal length of objective lens}$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (B) and (C) only
3. (A) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.17 In a magnetic field,

- (A) a charged particle in motion can experience a force
- (B) a charged particle at rest, experiences a force
- (C) there is no change in the speed of the charged particle
- (D) there is a change in the kinetic energy of the charged particle.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (C) only
3. (A), (B), and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.18 An LCR series circuit with $L = 100 \text{ mH}$, $C = 100 \mu\text{F}$, $R = 10 \Omega$ is connected to an AC source of emf 220 V. The resonant frequency of the circuit will be nearly

1. 50 Hz
2. 50 rad s^{-1}
3. 60 Hz
4. $50\pi \text{ Hz}$

Options 1. 1

2. 2
3. 3
4. 4

Q.19 A ray of light passes through an equilateral glass prism such that the angle of incidence (i) = angle of emergence (e).

If the angle of emergence is 0.75 times the angle of the prism, the refractive index of the prism will be

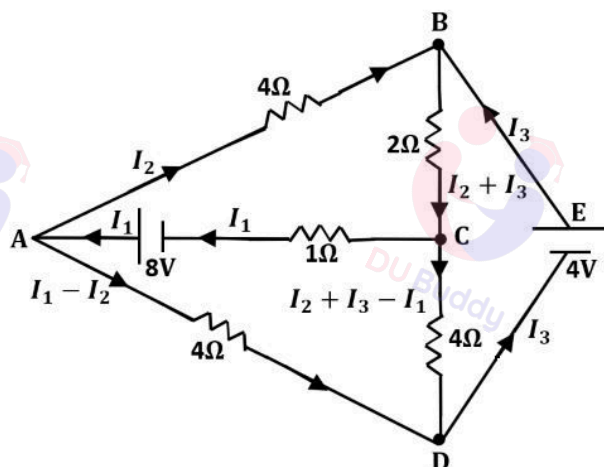
1. $2\sqrt{2}$
2. $\frac{1}{\sqrt{2}}$
3. $\sqrt{2}$
4. $\frac{1}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.20

Applying the Kirchhoff's rule to the loop (a) ABCA and (b) BCDEB, two correct equations are given



- (A) $8 - 4I_2 - 2(I_2 + I_3) - I_1 = 0$
 (B) $4 - 2(I_2 + I_3) - 4(I_2 + I_3 - I_1) = 0$
 (C) $4 + 2(I_2 - I_3) - 4(I_2 + I_3 - I_1) = 0$
 (D) $8 + 4I_2 + 2(I_2 - I_3) - I_1 = 0$

Choose the **correct** answer from the options given below:

1. (A) and (B) only
2. (B) and (C) only
3. (C) and (D) only
4. (D) and (A) only

Options 1. 1

2. 2
3. 3
4. 4

Q.21 For an AC circuit, a phase diagram shows emf on X -axis, and current I is at angle θ to emf. Angle θ is phase angle between emf and current. Current I is resolved into two components, $I \cos \theta$ is along emf on X -axis. $I \sin \theta$ is along perpendicular direction to emf.

- (A) Component $I \cos \theta$ will draw the power from the source
 (B) Component $I \sin \theta$ will draw the power from the source
 (C) $I \cos \theta$ is called the powerless current
 (D) $I \sin \theta$ is called the powerless current

Choose the correct answer from the options given below:

1. (B) and (C) only
2. (A) and (C) only
3. (C) and (D) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Identify the correct statements about the internal resistance of a cell

- (A) It depends on the nature of the electrolyte
 (B) It increases with the increase in temperature of the electrolyte
 (C) It increases with the decrease in temperature of the electrolyte
 (D) It increases with the distance between the two electrodes

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A), (B) and (D) only
3. (A), (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.23 A point source S is placed at a distance of 12 cm from a converging lens of focal length 10 cm on its principal axis. Where should a diverging mirror of focal length 14 cm be placed from the lens so that a real image is formed on the source itself?

1. 32 cm
2. 28 cm
3. 60 cm
4. 46 cm

Options 1. 1

2. 2
3. 3
4. 4

Q.24 A light of wavelength 500 nm falls on a metal surface, whose work function is 2.14 eV, The kinetic energy of the emitted electrons is

1. 3.4 eV
2. 0.34 eV
3. 2.48 eV
4. 0.034 eV

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.25 The resistance of a galvanometer is $G = 30\Omega$. It gives full scale deflection for $I_g = 2$ mA. It is to be converted into an ammeter to measure current $I = 0.3$ A. The value of the shunt required is approximately

1. 0.2 Ω
2. 2.03 Ω
3. 0.06 Ω
4. 0.09 Ω

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.26 A surface with area $\vec{S} = 20\hat{k}$ is held in an electric field $\vec{E} = \hat{i} + 2\hat{j} + 5\hat{k}$. The electric flux linked with the surface is

1. 160 units
2. 100 units
3. 60 units
4. 20 units

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.27 In a Young's double slit experiment, a light of wavelength 550 nm is used to get fringes on screen 1.2 m away from the slits. The distance between the two slits is 1.5 mm. The fringe width is

1. 4.4 mm
2. 0.44 mm
3. 5.5 mm
4. 4.0 mm

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.28 The reflected and the refracted rays are perpendicular to each other when light is incident on a transparent denser medium having refractive index $\mu = \sqrt{3}$. The angle of incidence is

1. 60°
2. 90°
3. 50°
4. 30°

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.29 An electrical circuit contains a battery of $V = 4$ V, and a resistance of $R = 2\Omega$. A galvanometer of resistance $G = 40\Omega$ shunted by $S = 10\Omega$ is used to measure current in the circuit.

1. 0.4 A
2. 2 A
3. 0 A
4. 0.1 A

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.30 Which of the following statements are correct?

- (A) The current under reverse bias is essentially voltage dependent on a critical reverse bias voltage.
- (B) The diode gets destroyed due to overheating if the forward or reverse current exceeds the rated value.
- (C) In forward bias measurement, we use a milliammeter since the expected current is large, while a microammeter is used in reverse bias to measure the current.
- (D) The drift current is of the order of a few μ A

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.31 In an experiment with Meter Bridge, with unknown resistance R in the left gap, the null point is obtained at $l_1 = 40$ cm from the left end. Now a resistance of $R' = 10\Omega$ is connected in series with resistance R , the null point is obtained at $l_2 = 70$ cm from the left end. The value of resistance R is

1. 4Ω
2. 6Ω
3. 5Ω
4. 3Ω

Options 1. 1

2. 2
3. 3
4. 4

Q.32 A wire of resistance $4R$ is bent in the form of a circle. The equivalent resistance between two points diametrically opposite on the circle will be

1. R
2. $2R$
3. $4R$
4. $R/4$

Options 1. 1

2. 2
3. 3
4. 4

Q.33 The electric field E associated with an electromagnetic wave is given by $E = E_0 \sin(kx - \omega t)$. The speed of the wave is

1. k
2. ω
3. ωk
4. $\frac{\omega}{k}$

Options 1. 1

2. 2
3. 3
4. 4

Q.34 In the observations of the experiment of scattering of Alpha-particle, Rutherford concluded about the atom that
(A) the positive charge and the mass of the atom are concentrated at the center called the nucleus
(B) there is positive charge centralized at one point of the size about 10^{-10} m inside the atom
(C) the atom being electrically neutral, it must have negatively charged electrons moving around the nucleus in different orbits
(D) the nucleus is small, electrons move around, there is no space left empty inside the atom
Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (A) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.35 A double convex lens of refractive index 1.5 has a radius of curvature R each. The focal length of the lens is

1. $R/2$
2. $2R$
3. $R/4$
4. R

Options 1. 1

2. 2
3. 3
4. 4

Q.36

Match List-I with List-II

List-I (Statement)	List-II (Energy band diagram)
(A) An intrinsic semiconductor at $T = 0\text{ K}$	(I)
(B) An intrinsic semiconductor at $T > 0\text{ K}$ with thermally generated electron-hole pairs.	(II)
(C) Energy bands of n-type semiconductor at $T > 0\text{ K}$	(III)
(D) Energy bands of p-type semiconductor at $T > 0\text{ K}$	(IV)

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Two particles carrying a charge of $q = 2\text{ nC}$ each connected by an insulating string are placed $r = 1\text{ m}$ apart on a smooth horizontal surface. The tension in the string is

1. $1.18 \times 10^{-8}\text{ N}$

2. $2.36 \times 10^{-8}\text{ N}$

3. $3.72 \times 10^{-8}\text{ N}$

4. zero

Options 1. 1

2. 2

3. 3

4. 4

Q.38 A light bulb is rated as 100 W , 230 V and is connected to a 230 V ac source. The value of peak current is

1. 0.61 A

2. 0.3 A

3. 0.39 A

4. 0.43 A

Options 1. 1

2. 2

3. 3

4. 4

Q.39 An ammeter and a Voltmeter are to be connected in a circuit. Which of the following statements are correct?

- (A) Ammeter has negligible resistance.
- (B) Ammeter is always connected in series in the circuit.
- (C) Voltmeter should have negligible resistance.
- (D) Voltmeter should be connected in parallel so that the whole current may pass through it.

Choose the correct answer from the options given below:

- 1. (A) and (B) only
- 2. (B) and (C) only
- 3. (A) and (D) only
- 4. (B) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.40 9 cells each of emf $e = 3\text{ V}$ and internal resistance $r = 1\Omega$ are connected in $m = 3$ rows, each having $n = 3$ cells in series.

The current flowing in the external resistance of $R = 5\Omega$ connected across this arrangement of the cells is

- 1. 1.2 A
- 2. 1.76 A
- 3. 1.5 A
- 4. 3 A

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.41 Which of the following are the properties of photons?

- (A) The rest mass of a photon is zero
- (B) Each photon has a definite energy and an indefinite linear momentum
- (C) A photon may collide with a material particle and the total energy and the total momentum remain conserved in such a collision
- (D) Photons are deflected by electric and magnetic fields

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (B) and (C) only
- 3. (A) and (C) only
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.42 In forward bias and in reverse bias, respectively, the width of the depletion layer of a p-n junction

- 1. decreases; increases
- 2. increases; decreases
- 3. decreases; decreases
- 4. increases; increases

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.43 The following have zero average value in a plane electromagnetic wave

- 1. electric field only
- 2. magnetic field only
- 3. both electric field and magnetic field
- 4. both electric energy and magnetic energy

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.44 If an electron and a photon have energy, then,

- (A) Electrons and photons both have wavelengths
- (B) Electron and photon both have momentum
- (C) Both electron and photon do not have momenta
- (D) The electron has energy and momentum and the photon has energy only

Choose the correct answer from the options given below:

- 1. (A) and (C) only
- 2. (A) and (B) only
- 3. (B) and (C) only
- 4. (A) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.45 Which of the following statements are correct?

- (A) The resistance of a conductor and its resistivity both depend on temperature.
- (B) For a conductor of 1 m length and of 1 m^2 cross-sectional area, the resistance is equal to its resistivity
- (C) Resistance and resistivity both depend on cross-sectional area
- (D) The resistivity of two wires made of same material but of different dimensions is the same.

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (D) and (C) only
- 3. (B), (D) and (C) only
- 4. (A) and (C) only

Options 1. 1

- 2. 2
- 3. 3

- Q.46** One student has to make an electromagnet using a coil of copper wire. He needs a ferromagnetic material to make its core. This core material should have
1. Low permeability, low retentivity
 2. High permeability, high retentivity
 3. Low permeability, high retentivity
 4. High permeability, low retentivity

Options 1. 1
2. 2
3. 3
4. 4

- Q.47** Three charges, $+Q$, q and $+Q$ are placed with q at mid point of the line joining the other two charges. For the system of charges to be in equilibrium, the value of q is

1. $+Q/4$
2. $-Q/4$
3. $+Q/2$
4. $-Q/2$

Options 1. 1
2. 2
3. 3
4. 4

- Q.48** A wire of $R_{\text{original}} = 10\Omega$ resistance is stretched to thrice its original length. Its new resistance R_{new} would be

1. 10Ω
2. 3Ω
3. 90Ω
4. 30Ω

Options 1. 1
2. 2
3. 3
4. 4

- Q.49** A small charged water drop with charge $q = 1.5 \times 10^{-6} \text{ C}$ and mass $m = 10 \text{ g}$ is suspended in air in electric field E . The value of E is given by (Take acceleration due to gravity $g = 10 \text{ m s}^{-2}$)

1. $10.2 \times 10^6 \text{ N/C}$
2. $1.5 \times 10^6 \text{ N/C}$
3. $1.5 \times 10^{-4} \text{ N/C}$
4. $6.67 \times 10^4 \text{ N/C}$

Options 1. 1
2. 2
3. 3
4. 4

- Q.50** A girl peddles a stationary bicycle. The pedals of the bicycle are attached to a $N = 100$ turn coil of area $A = 0.10 \text{ m}^2$. The coil completes one revolution in $T = 2$ seconds, and it is placed in a uniform magnetic field of $B = 0.02 \text{ T}$ perpendicular to the axis of rotation of the coil. The maximum voltage generated in the coil will be

1. 0.314 V
2. 3.14 V
3. 6.28 V
4. 628 V

Options 1. 1
2. 2
3. 3
4. 4